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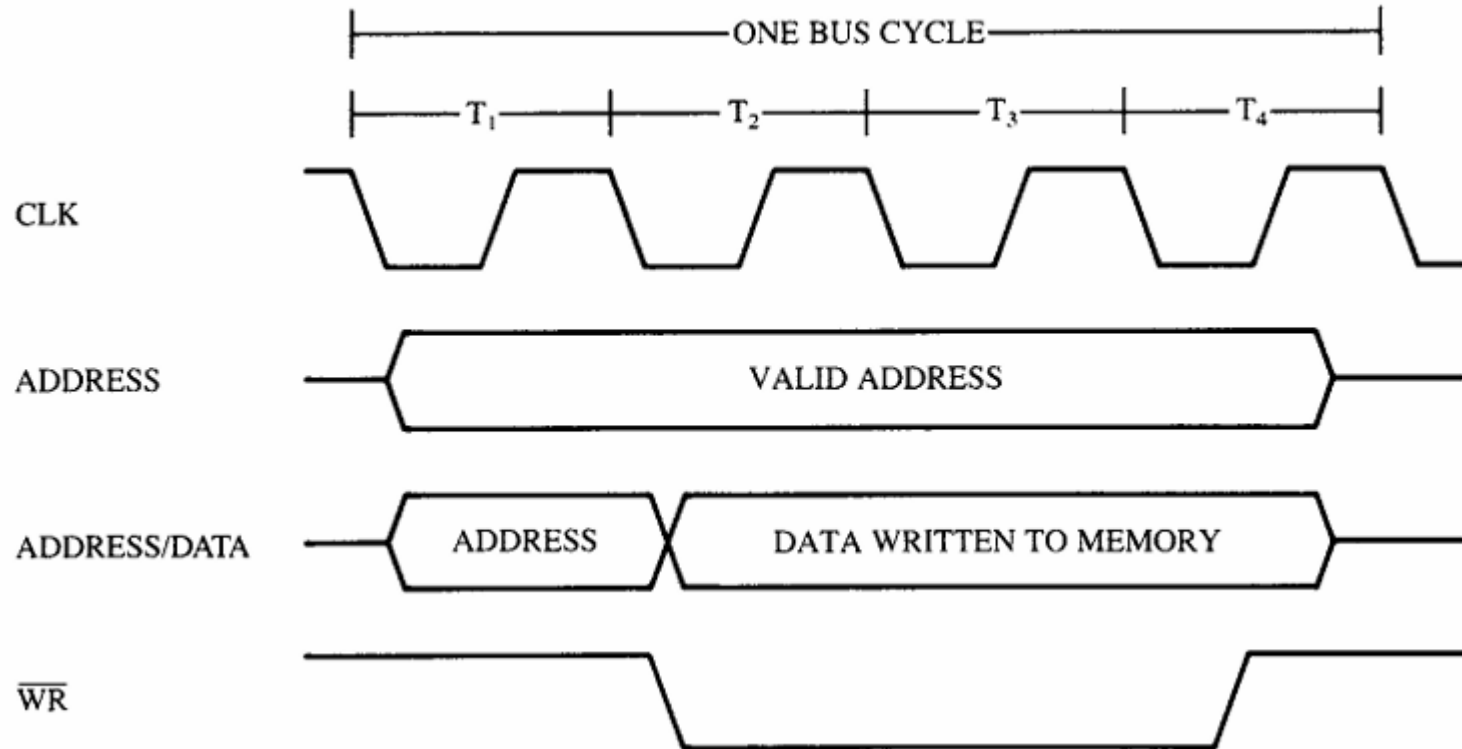
8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

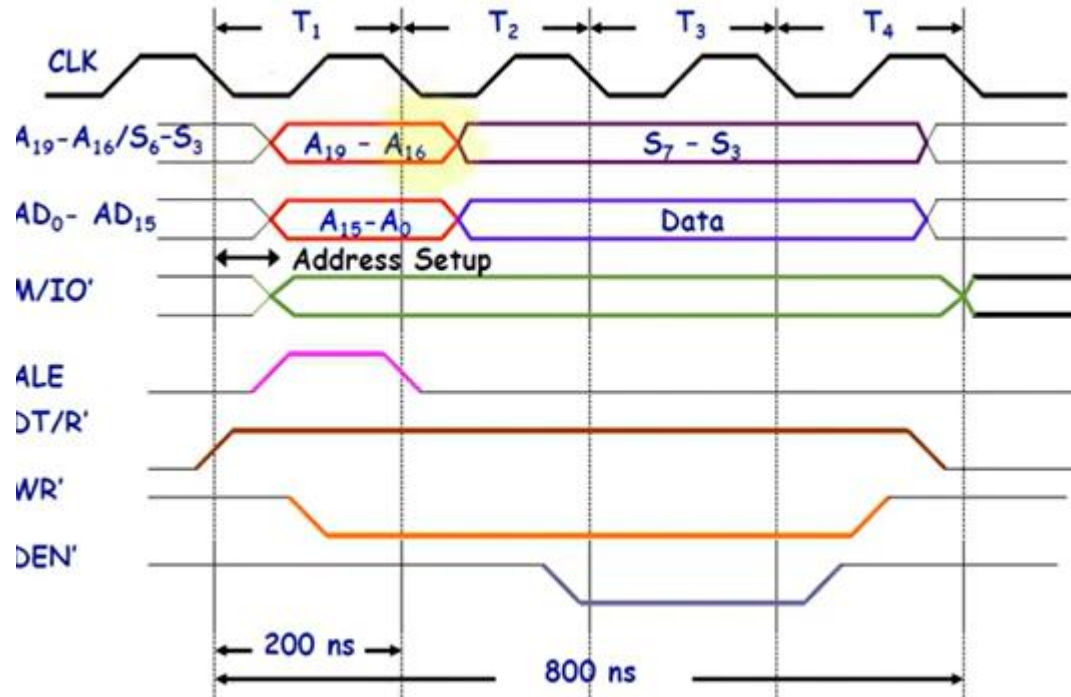
Dr. Manideepa Mukherjee
Department of CS & IS



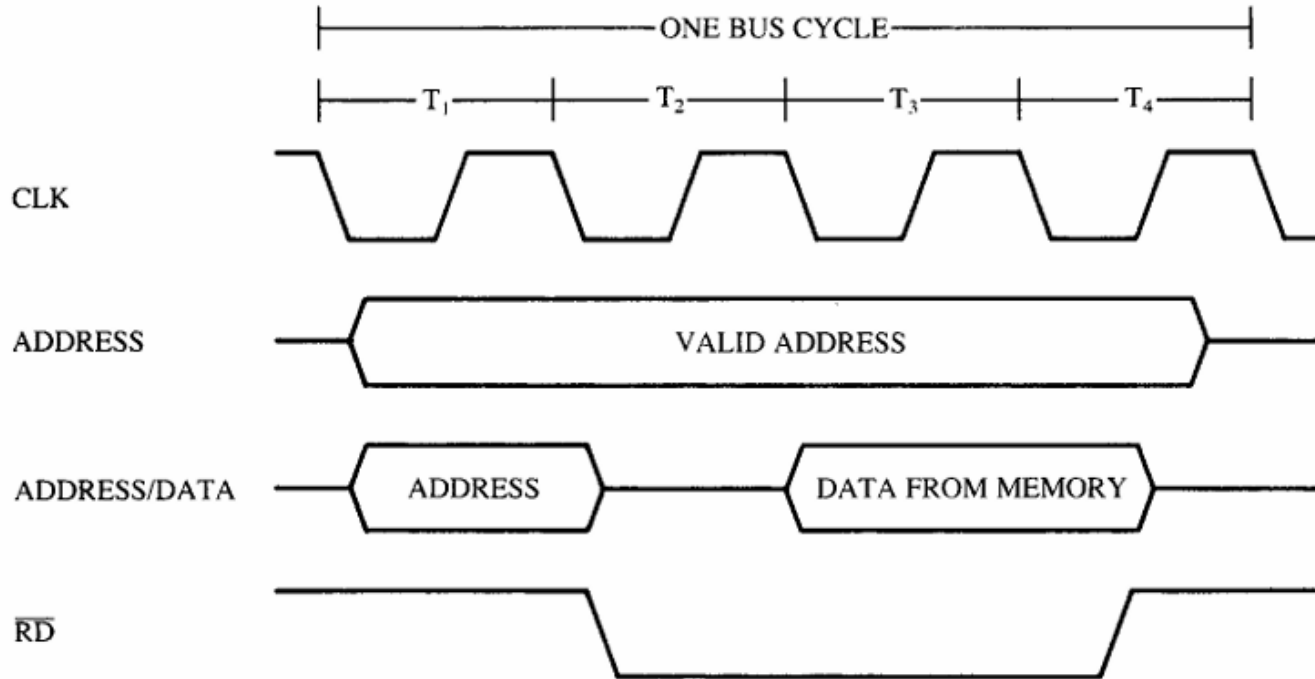
Timing Diagram Write operation



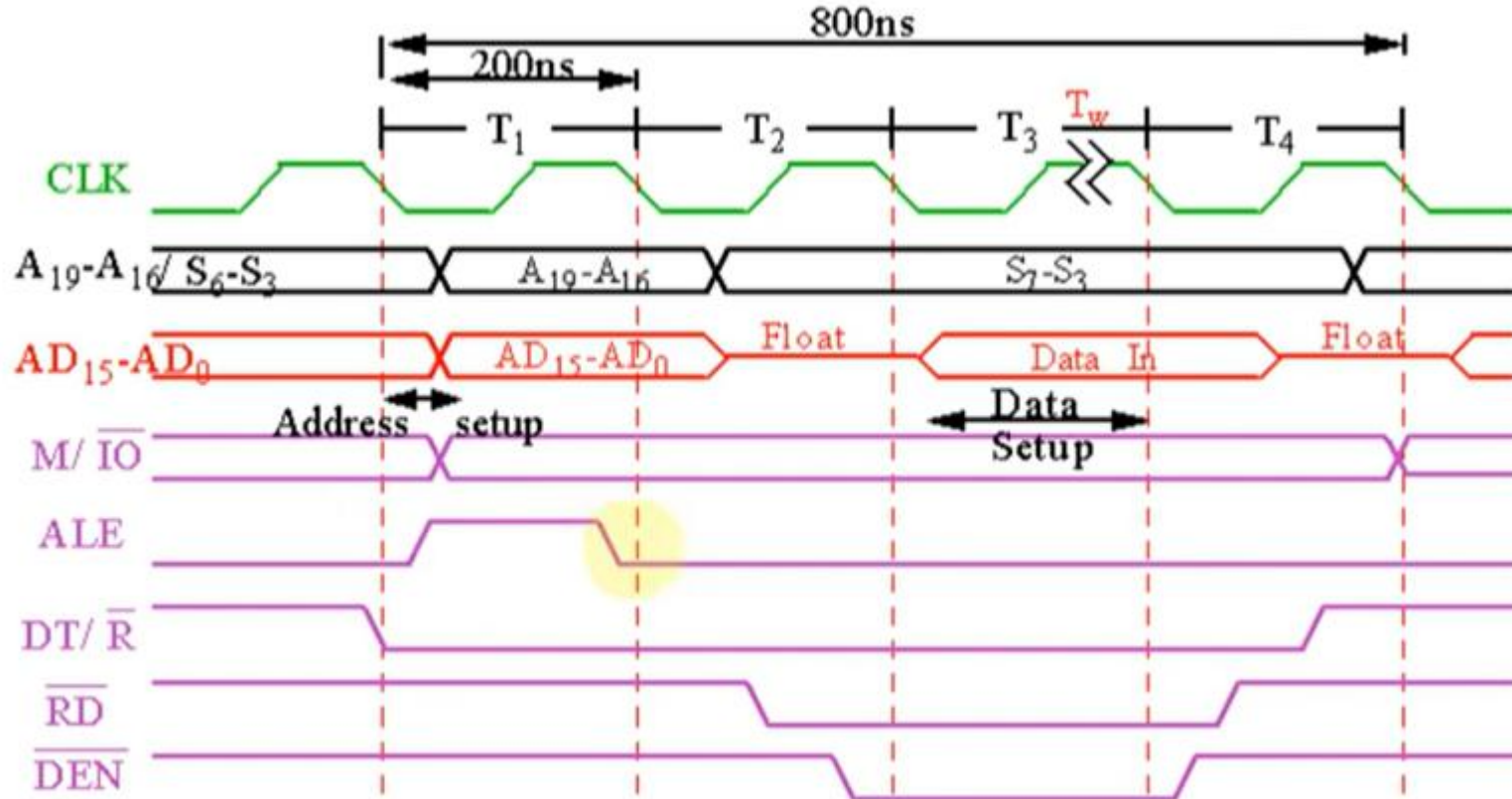
Memory Write Operation



Timing Diagram Read operation

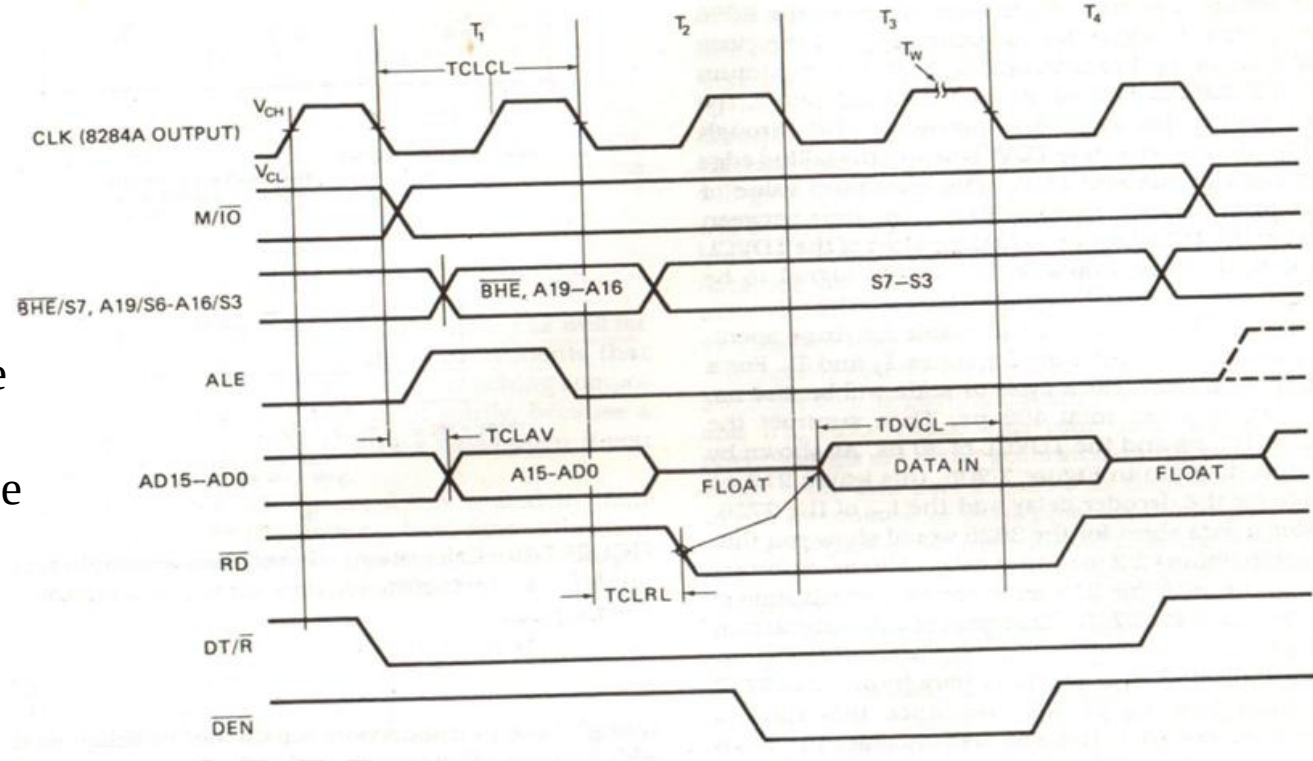


Memory Read Operation



Memory Access Time

TCLAV- Time from Clock to Address Valid ;
110 ns for 5 MHz Clock
Memory Access Time :
TCLAV time must be subtracted from the three clocking states (600 ns) separating the appearance of the address (T_1) and the sampling of the data (T_3).



Memory Access Time

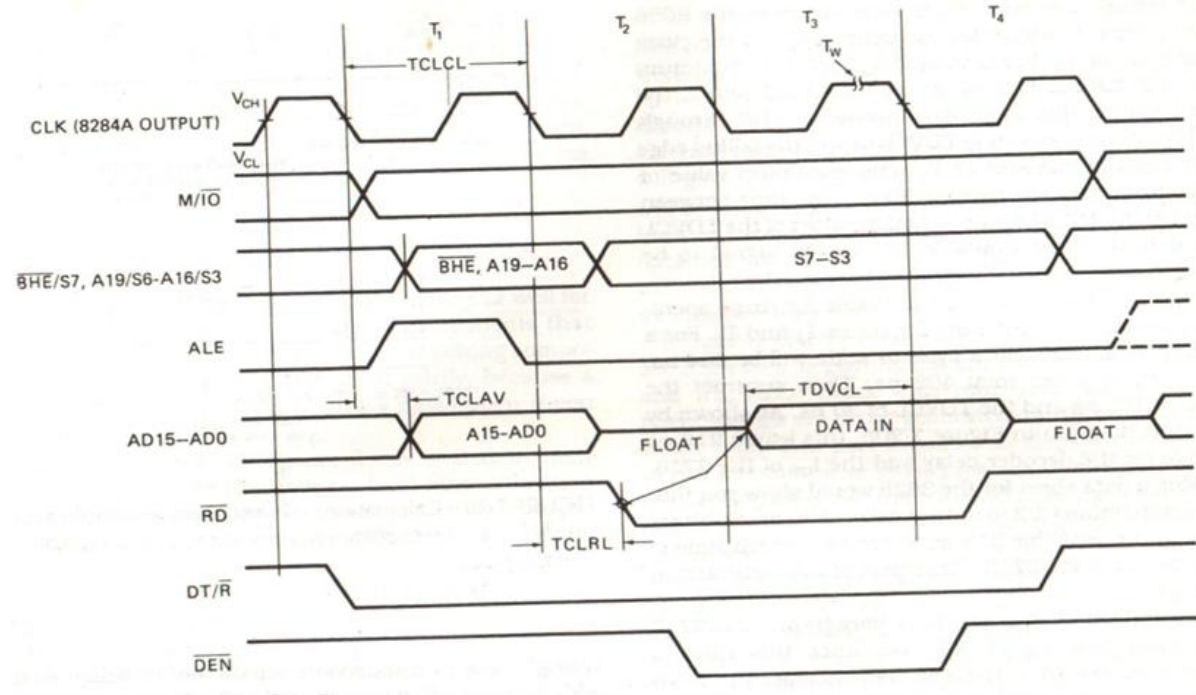
TCLRL-Time from Clock to Read Line

TDVCL –Time Data Valid to Clock 30ns

Memory Access Time : 600 ns-
110ns-30ns = 460 ns

A **wait state** (T_w) is an extra clocking period between T_2 and T_3 to lengthen bus cycle

On one wait state, memory access time of 460 ns, is lengthened by one clocking period (200 ns) to 660 ns, based on a 5 MHz clock.



Problem1: An 8086 operates at 8MHz (8086-2). If one wait state (T_w) is inserted, what is the total time allowed for memory access?

Problem2: A processor is working in 8 MHz and memory access time is 300 ns. considering address setup time of 120 ns and data setup time 20 ns with a latching buffer delays of 10 ns. What will be the time taken for reading 16 bit data stored in memory location 2010H

Memory Types

- Two basic types:
- ROM: Read-only memory
- RAM: Read-Write memory

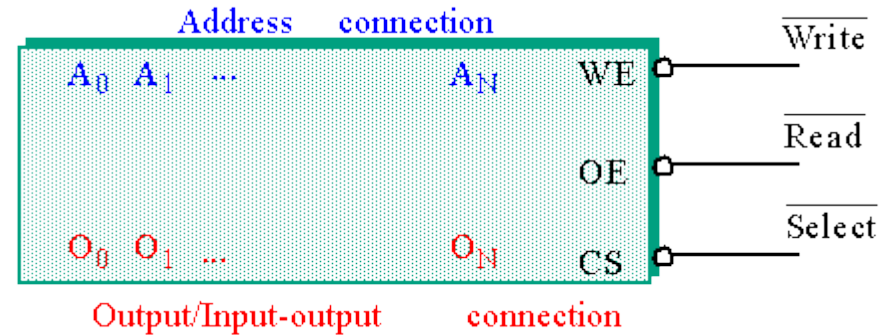
Four commonly used memories:

ROM

Flash (EEPROM)

Static RAM (SRAM)

Dynamic RAM (DRAM)



Memory Chips

The number of **address pins** is related to the number of **memory locations**.

Common sizes today are **1K to 256M** locations.

Therefore, between **10 and 28 address pins** are present.

The data pins are typically bi-directional in read-write memories.

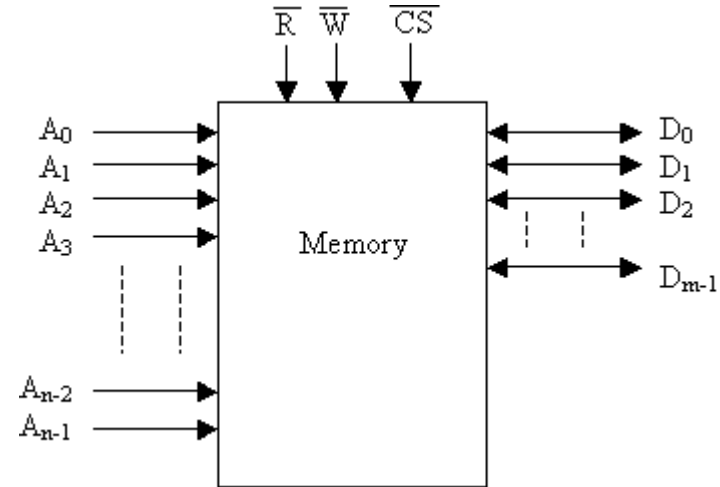
The number of data pins is related to the size of the memory location.

For example, an **8-bit wide (byte-wide)** memory device has **8 data pins**.

Catalog listing of **1K X 8** indicate a byte addressable **8K** memory.

Memory Chips

- Each memory device has at least one chip select (**CS**) or chip enable (**CE**) or select (**S**) pin that enables the memory device.
- This enables **read and/or write** operations. If more than one are present, then all must be **0** in order to perform a read or write.



Memory chips

Each memory device has at least one control pin.

For **ROMs**, an output enable (**OE**) or gate (**G**) is present.

The **OE pin enables** and disables a set of tristate buffers.

For **RAMs**, a read-write (**R/W**) or write enable (**WE**) and read enable (**OE**) are present.

For dual control pin devices, it must be hold true that both are not 0 at the same time.

ROM

Non-volatile memory: Maintains its state when powered down.

There are several forms:

ROM: Factory programmed, cannot be changed. Older style.

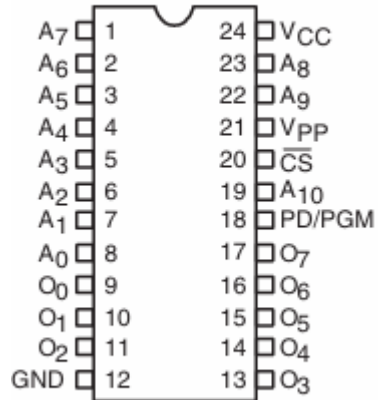
PROM: Programmable Read-Only Memory. Field programmable but only once. Older style.

EPROM: Erasable Programmable Read-Only Memory. Reprogramming requires up to 20 minutes of high-intensity UV light exposure.

Flash EEPROM: Electrically Erasable Programmable ROM. Also called EAROM (Electrically Alterable ROM) and NOVRAM (Non-Volatile RAM). Writing is much slower than a normal RAM.

ROM (2716, 2K X 8 EPROM)

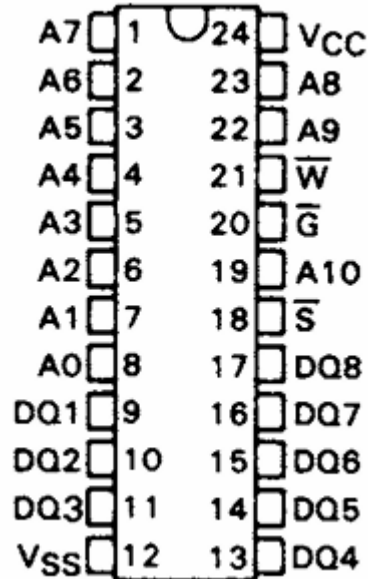
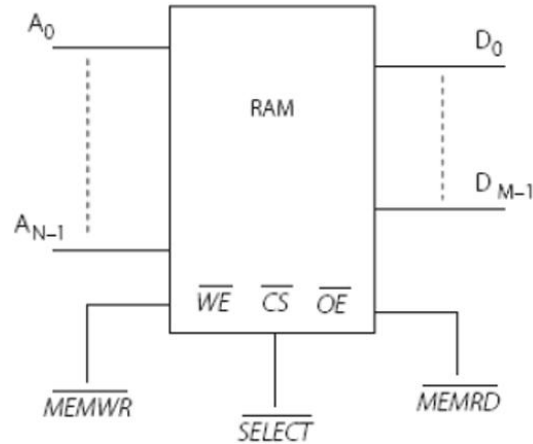
PIN CONFIGURATION



PIN NAMES

A ₀ –A ₁₀	ADDRESSES
PD/PGM	POWER DOWN/PROGRAM
$\overline{\text{CS}}$	CHIP SELECT
O ₀ –O ₇	OUTPUTS

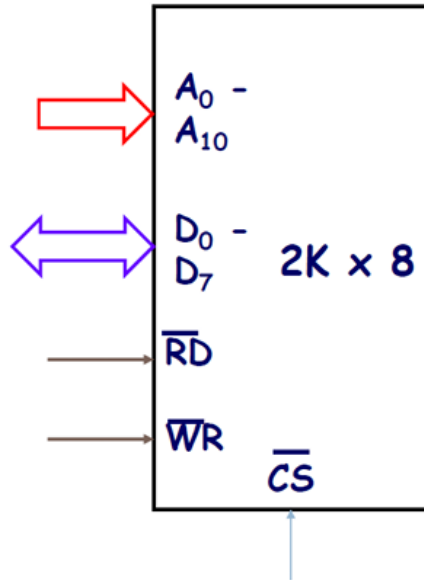
Interface RAM



PIN NOMENCLATURE	
A0 – A10	Addresses
DQ1 – DQ8	Data In/Data Out
$\overline{G}$	Output Enable
$\overline{S}$	Chip Select
V_{CC}	+5-V Supply
V_{SS}	Ground
$\overline{W}$	Write Enable

Memory Chip

A 2K Memory Chip



Memory Address Decoding

The processor can **usually address a memory space** that is **much larger than the memory** space covered by an individual memory chip.

In order to **splice a memory device** into the **address space of the processor**, decoding is necessary.

For example, the **8086 issues 20-bit addresses** for a total of **1MB of** memory address space.

However, the BIOS on a 2716 EPROM has **only 2KB of** memory and **11 address pins**.

A decoder can be used to decode the additional **9 address pins** and allow the **EPROM to be placed in any 2KB section of the 1MB address space**.

Interfacing Memory

In general to address a memory location out of N memory locations, we will require at least n bits of address, i.e. n address lines where $n = \log_2 N$.

Thus if the microprocessor has n address lines, then it is able to address at the most N locations of memory, where $2^n = N$.

If out of N locations only P memory locations are to be interfaced, then the least significant p address lines out of the available n lines can be directly connected from the microprocessor to the memory chip while the remaining $(n-p)$ higher order address lines may be used for address decoding as inputs to the chip selection logic.

Address Decoding

1111 1111 1XXX XXXX XXXX

or

1111 1111 1000 0000 0000 = FF800H

to

1111 1111 1111 1111 1111 = FFFFFH

Memory
2K X 8

CS'

Select'

A11

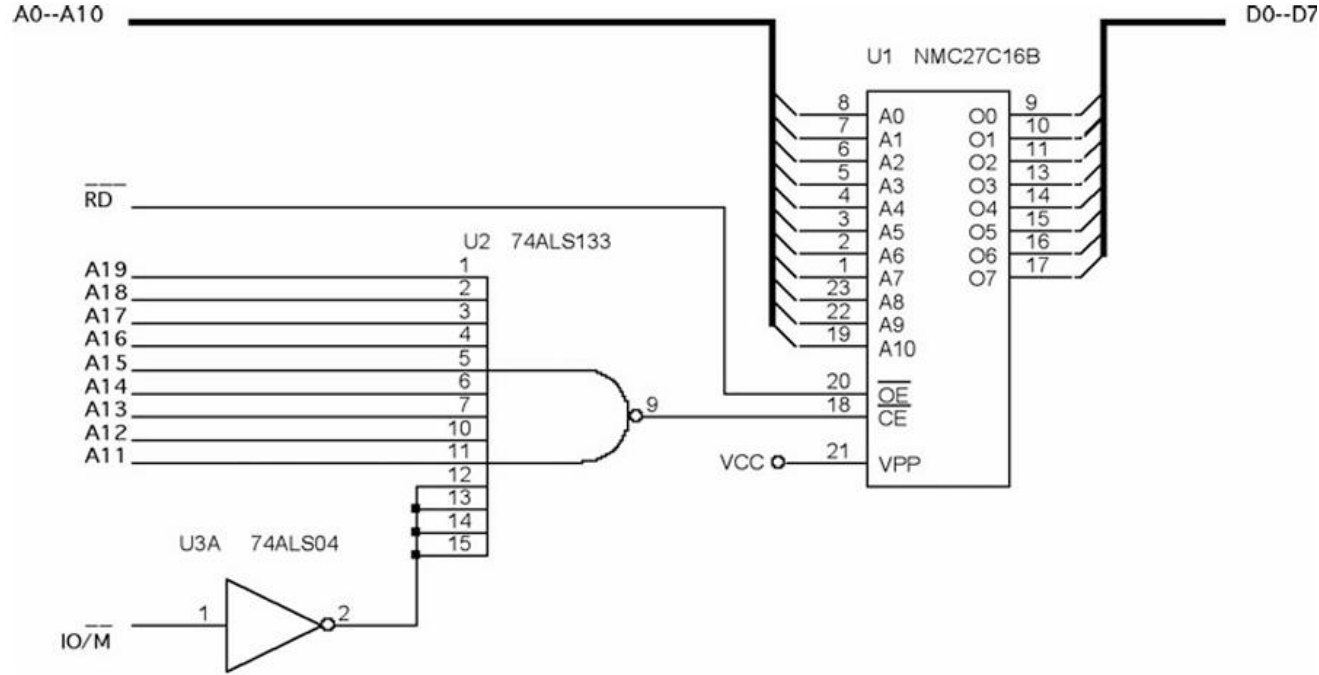
A19

=FF800

=FFFF

A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
1	1	1	1	1	1	1	1	1	1	9	8	7	6	5	4	3	2	1
9	8	7	6	5	4	3	2	1	0									
1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

NAND Decoder



Example

Modify the NAND gate decoder to select the memory for address range DF800H–DFFFFH.

DF800H = 1101 1111 1000 0000 0000

DFFFFH = 1101 1111 1111 1111 1111